Where Does Label Noise Hurt Ensembles? A 0-1 Bias–Variance Decomposition of Bagging, Random Forests and Boosting under Symmetric, One-Sided and Boundary Label Noise

Krish Kalola
Department of Computer Science
Pandit Deendayal Energy University
Gandhinagar, Gujarat, India
krishkalola@outlook.com

Harshi Jain
Department of Computer Science
Pandit Deendayal Energy University
Gandhinagar, Gujarat, India
harshi.r.jain2005@gmail.com

Pearl Kalariya
Department of Computer Science
Pandit Deendayal Energy University
Gandhinagar, Gujarat, India
pearlkalariyaofficial@gmail.com

***Abstract*—Tree ensembles are the default choice for tabular classification, yet real training labels often contain errors. Earlier studies report how much accuracy ensembles lose under label noise, but not whether the extra error comes from learning a wrong decision rule (bias) or from instability (variance), and they mostly flip labels at random. We measure this directly. On 15 binary datasets selected by a fixed rule from the OpenML-CC18 benchmark, we inject symmetric, one-sided and boundary label noise at rates up to 30 percent and decompose the clean-test error of a decision tree, bagging, random forest, AdaBoost, gradient boosting and XGBoost into bias and variance with the zero-one loss decomposition of Domingos. Symmetric noise adds almost only variance. One-sided noise adds mainly bias to shallow boosted models but mainly variance to deep trees, so the type of damage depends on both the noise and the model capacity. Boundary noise splits roughly evenly and concentrates its damage near the decision boundary. Boosting appears more robust than bagging under default settings, but a capacity-controlled experiment shows that this is a tree-depth effect: at equal depth, bagging and random forests are significantly more robust. Variance-reducing remedies such as small-sample bagging and early stopping recover up to 56 percent of the damage from symmetric noise but no significant share of the damage from one-sided noise. A check on 20 multiclass datasets confirms the patterns for symmetric and boundary noise. Matching the remedy to the noise type therefore matters more than the choice of ensemble.**

***Index Terms*—label noise, ensemble learning, bias–variance decomposition, bagging, boosting, random forest**

I. Introduction

Tree-based ensembles such as bagging [1], random forests [2] and boosting [3]–[5] remain the strongest general-purpose models for tabular data. In practice their training labels are rarely perfect: annotators make mistakes, ambiguous cases are labelled inconsistently and automated labelling introduces systematic errors [6]. It is therefore important to know how label noise affects each kind of ensemble.

Much is already known about *how much* noise hurts. Dietterich [7] showed that bagging is more robust than boosting when labels are flipped at random, and recent work shows that the robustness ranking of modern gradient-boosting libraries depends on the type of noise [8]. These studies, however, report only accuracy-type degradation. They do not say *why* an ensemble degrades: does noise make the model learn a systematically wrong rule (bias), or does it make the model unstable (variance)? The answer is practically important, because the two problems need opposite remedies. Variance can be reduced by averaging or early stopping; bias caused by wrong labels can only be removed by correcting the data.

This paper measures the answer. We combine controlled label-noise injection with the zero-one loss bias–variance decomposition of Domingos [9] and ask four questions:

- **RQ1:** How does each ensemble's error grow with the noise rate?
- **RQ2:** Is the extra error mainly bias or variance, and does this depend on the noise type and the ensemble family?
- **RQ3:** Does the noise-induced variance come from *which* labels are flipped or from *which* rows are sampled?
- **RQ4:** Do standard regularisation remedies recover the damage under every noise type?
- **RQ5:** Do the binary findings still hold for multiclass data?

Our contributions are: (i) the first measured bias–variance attribution of noise-induced error across parallel and sequential tree ensembles under symmetric, one-sided and boundary noise; (ii) evidence for one-sided class-conditional noise on binary tasks, a setting left open by recent work [8]; (iii) a capacity-controlled experiment showing that the apparent robustness of boosting under default settings is a tree-depth effect; (iv) practical guidance that matches remedies to noise types; and (v) a check on 20 multiclass datasets showing which findings generalise beyond binary tasks. All code and results are released for reproducibility [TODO: GitHub link].

II. Background

A. Tree Ensembles

Bagging trains each tree on a bootstrap sample and combines them by majority vote [1]; random forests additionally sample features at each split [2]. Both are *parallel* ensembles whose main effect is variance reduction. AdaBoost [3] and gradient boosting [4], [5] are *sequential*: each new tree concentrates on the examples that earlier trees got wrong, which mainly reduces bias but can also make the ensemble chase mislabelled examples [10].

B. Label-Noise Types

Following the taxonomy of Frénay and Verleysen [6], we study three noise types. *Symmetric* noise flips labels completely at random (NCAR). *One-sided* noise flips only class 1 to class 0, a class-conditional model (NAR) [11] that describes, for example, missed diagnoses. *Boundary* noise flips mainly examples near the decision boundary, an instance-dependent model (NNAR) that reflects human errors on ambiguous cases [12].

C. Zero-One Loss Bias–Variance Decomposition

Let a learner be trained R times on different training samples and let $\hat{y}_r(x)$ be the prediction of the r-th model for a test point x with true label y. Following [9], the *main prediction* $y_m(x)$ is the majority vote of the R models, the bias is $B(x) = \mathbb{1}[y_m(x) \neq y]$ and the variance is $V(x) = \Pr_r[\hat{y}_r(x) \neq y_m(x)]$. For binary labels and noise-free test labels the expected loss is

$$\mathbb{E}[L(x)] = B(x) + \big(1 - 2B(x)\big)\, V(x). \tag{1}$$

Variance therefore *adds* to the error where the main prediction is correct (unbiased variance) and *subtracts* from it where the main prediction is wrong (biased variance). Averaged over the test set,

$$\text{error} = \text{bias} + \underbrace{V_u - V_b}_{\text{net variance}}, \tag{2}$$

where V_u and V_b are the average unbiased and biased variance. We report the bias and the net variance, which add up exactly to the error.

III. Related Work

Dietterich [7] compared bagging, boosting and randomisation of C4.5 trees and found bagging much better under random classification noise. Khoshgoftaar *et al.* [13] reached a similar conclusion for noisy, imbalanced binary data. Long and Servedio [10] proved that random classification noise can defeat any convex-potential booster, even with early stopping. Sabzevari *et al.* [14] showed that bagging with small bootstrap samples is more robust to label noise, and Sáez and Romero-Béjar [12] studied bagging under borderline noise. Eisenbürger *et al.* [15] studied noise detection for gradient-boosted trees, and Morampudi *et al.* [8] showed that the robustness ranking of XGBoost, LightGBM and CatBoost depends on the noise model; they measured only accuracy-type metrics and explicitly left one-sided binary noise and remedies under non-symmetric noise open. On the theory side, Wood *et al.* [16] unified bias, variance and diversity in ensembles, and He *et al.* [17] used this framework to design noise-robust ensemble algorithms. Mohammad [18] discussed ensembles on tabular data from a bias–variance perspective but without injected noise or a formal decomposition. To our knowledge no study *measures* how noise-induced error splits into bias and variance across noise types and ensemble families. [TODO: After reading the full text of He et al. (2026), confirm this last sentence.]

TABLE I
Datasets (OpenML-CC18), ordered by size. Pos. is the share of class 1.

Dataset	ID	Rows	Feat.	Pos.
kc2	1063	522	21	0.20
wdbc	1510	569	30	0.37
ilpd	1480	583	11	0.29
blood-transfusion	1464	748	4	0.24
diabetes	37	768	8	0.35
tic-tac-toe	50	958	27	0.35
credit-g	31	1,000	61	0.30
qsar-biodeg	1494	1,055	41	0.34
banknote-authentication	1462	1,372	4	0.45
kc1	1067	2,109	21	0.15
kr-vs-kp	3	3,196	73	0.48
spambase	44	4,601	57	0.39
phoneme	1489	5,404	5	0.29
PhishingWebsites	4534	11,055	68	0.44
electricity	151	45,312	14	0.42

IV. Experimental Setup

A. Datasets

To avoid hand-picking, we used *all* binary datasets of the OpenML-CC18 benchmark [19], [20] with at most 50,000 rows, at most 100 features, no missing values and a minority class of at least 15%. Of the 35 binary CC18 datasets, 15 satisfy the rule (Table I). Labels were coded so that the minority class is class 1. Each dataset was split once into a stratified 70% training pool and a 30% test set; categorical features were one-hot encoded using the training part only. Label noise was injected into training labels only, so all errors are measured against clean test labels.

B. Models

We compared six classifiers implemented in scikit-learn [21] and XGBoost [5]: an unpruned decision tree; bagging and a random forest, each with 100 unpruned trees; AdaBoost with 100 decision stumps; gradient boosting (GB) with 100 trees of depth 3 and learning rate 0.1; and XGBoost (XGB) with 100 trees of depth 4 and learning rate 0.1. Hyperparameters were fixed and identical in every noise condition, so that changes in error are caused by the noise alone.

C. Noise Injection

For a rate ρ, symmetric noise flips exactly ρn labels chosen uniformly at random. One-sided noise flips exactly ρn_1 labels

of class 1 to class 0, where n_1 is the number of positives. Boundary noise flips exactly ρn labels, choosing each example with probability proportional to $1 - 2|p(x) - 0.5|$, where $p(x)$ is the class-1 probability of a logistic regression trained on the clean labels. We used $\rho \in \{5, 10, 20, 30\}\%$.

D. Protocol

For every dataset, model, noise type and rate, the model was trained $R = 51$ times, each time on a fresh bootstrap sample of the training pool with freshly injected noise, and evaluated on the clean test set. An odd R avoids ties in the majority vote. The bootstrap sample and noise mask of round r depend only on the round, not on the model, so all models face identical training data (common random numbers), and every comparison between models is paired. The main grid has 15 datasets, 6 models and 13 noise conditions (clean, and three noise types at four rates), giving 1,170 conditions and about 60,000 model fits. The *damage* caused by noise is the change in error, bias and net variance relative to the same model on clean labels; by construction $\Delta\text{error} = \Delta\text{bias} + \Delta\text{net variance}$.

E. Additional Experiments

Three further experiments reuse the same datasets, seeds and protocol at rates of 10, 20 and 30%.

Capacity control. Under default settings the parallel ensembles grow full-depth trees, whereas AdaBoost uses stumps and GB and XGB use trees of depth 3 and 4. To separate the ensemble method from tree size, we added depth-matched variants: bagging, random forest, AdaBoost and XGB with depth-3 trees, and GB and XGB with depth-8 trees.

Sources of variance (RQ3). At 20% noise, each model was additionally trained under three designs that re-draw only part of the randomness: only the model's own random seed (training rows and noise mask fixed), only the noise mask (training rows fixed) and only the bootstrap sample (noise mask fixed). Subtracting the variance of the first design from the other two gives the variance added by the noise realisation and by sampling.

Remedies (RQ4). Each of four standard remedies was compared with its base model: GB and XGB with learning rate 0.05, 80% row subsampling and early stopping on a 10% split of the (noisy) training data; AdaBoost with learning rate 0.5; and bagging with bootstrap samples of 20% of the training set [14]. The *damage recovered* is $1 - \Delta\text{error}_{\text{remedy}}/\Delta\text{error}_{\text{base}}$.

Finally, to see *where* the damage lands, each test set was split at the median of the margin $|p(x) - 0.5|$ of the clean logistic-regression reference into a near-boundary and a far-from-boundary half, and the saved predictions of the main grid were decomposed separately in each half.

F. Multiclass Extension (RQ5)

Applying the same selection rule to the multiclass datasets of OpenML-CC18 (with a smallest class of at least 5%) yields 20 datasets with 3 to 11 classes, prepared exactly as the binary ones. Symmetric noise moves a flipped label to a uniformly chosen other class. As the multiclass counterpart of one-sided noise we use *pair-flip* noise, which moves class k to class $k + 1$ (classes ordered by frequency). Boundary noise selects examples with probability proportional to one minus the top-class probability of a multinomial logistic-regression reference and moves each to the reference's most likely other class. With $K > 2$ classes, a model that disagrees with a wrong main prediction lowers the error only if it predicts the true class, so the biased variance in the decomposition is replaced by the probability of predicting the true class at biased points; the identity error = bias + net variance still holds exactly and reduces to (1) for $K = 2$. Rates of 10, 20 and 30% and the same six models and protocol were used.

TABLE II
AVERAGE RANK OF DAMAGE AT 20% NOISE OVER 15 DATASETS (1 = LEAST DAMAGED)

Model	Sym.	One-s.	Bound.
Tree	5.60	3.53	5.27
Bagging	3.87	2.73	4.13
RF	3.40	4.07	3.73
AdaBoost	1.73	3.67	1.33
GB	3.13	3.73	2.80
XGB	3.27	3.27	3.73
Friedman p	2.7e-06	0.482	4.2e-07

G. Statistical Analysis

Following Demšar [22], models were compared over the 15 datasets with the Friedman test on their ranks, followed by the Nemenyi critical difference (CD) at $\alpha = 0.05$; pairs of models were compared with the Wilcoxon signed-rank test.

V. RESULTS

A. RQ1: How Much Does Noise Hurt?

Fig. 1 shows the average increase in test error. The single tree is hurt most under every noise type. With default settings the sequential ensembles lose less accuracy than the parallel ones under symmetric and boundary noise: at 30% symmetric noise the error of bagging rises by 11.7 percentage points (pp) and that of random forest by 10.9 pp, against 4.3 pp for AdaBoost and 7.6 pp for GB. At equal rate, one-sided noise is the mildest type for every model except AdaBoost, which is hurt slightly more by one-sided than by symmetric noise.

Table II ranks the models by damage on each dataset. Under symmetric and boundary noise the differences are highly significant (Friedman $p < 10^{-5}$). AdaBoost has the best average rank, and with a CD of 1.95 its advantage over the single tree and bagging is significant. Under one-sided noise, in contrast, no model is consistently more robust ($p = 0.48$): the choice of ensemble does not protect against one-sided noise. Section V-C shows that the advantage of boosting under the other two noise types is explained by tree depth.

B. RQ2: Bias or Variance?

Fig. 2 and Table III split the damage at 20% noise. Under *symmetric* noise the damage is almost pure variance (96–105% of the damage for every model except AdaBoost, 71%):

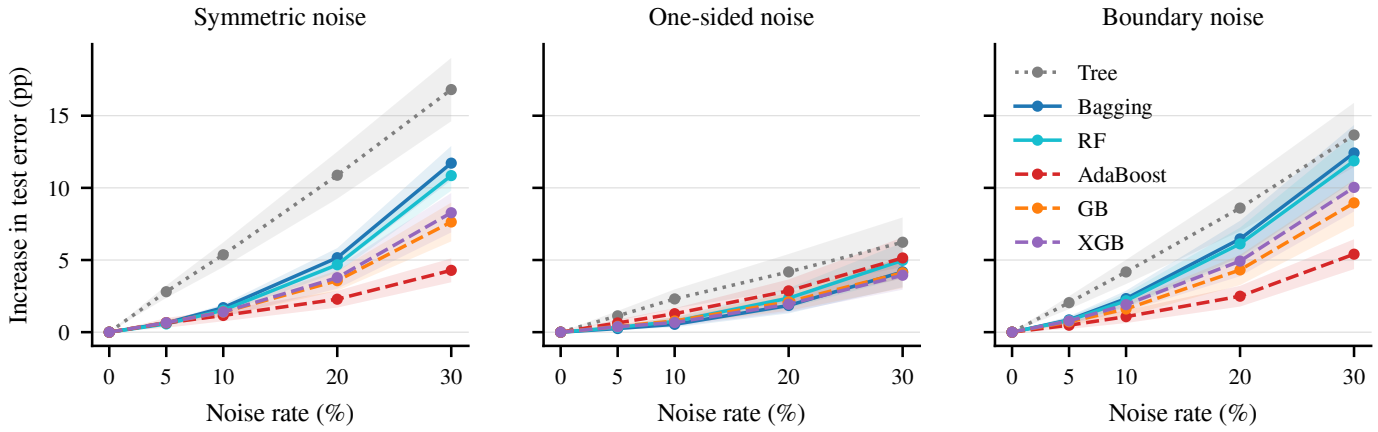


Fig. 1. Increase in clean-test error caused by label noise, averaged over 15 datasets (shaded: ±1 standard error).

TABLE III
DAMAGE AT 20% NOISE (PP) AND THE SHARE DUE TO VARIANCE

Model	Δerr	Δbias	Δvar	Var. share
Symmetric noise				
Tree	10.9	−0.1	11.0	101%
Bagging	5.2	−0.3	5.4	105%
RF	4.7	0.1	4.6	99%
AdaBoost	2.3	0.7	1.6	71%
GB	3.6	0.1	3.5	98%
XGB	3.8	0.2	3.6	96%
One-sided noise				
Tree	4.2	0.2	3.9	94%
Bagging	1.8	0.2	1.6	89%
RF	2.4	1.5	0.9	36%
AdaBoost	2.9	2.6	0.2	7%
GB	2.1	1.8	0.3	17%
XGB	1.9	1.2	0.8	40%
Boundary noise				
Tree	8.6	3.9	4.7	54%
Bagging	6.5	3.3	3.2	49%
RF	6.1	3.4	2.7	44%
AdaBoost	2.5	1.6	0.9	37%
GB	4.3	2.6	1.7	40%
XGB	4.9	3.0	1.9	39%

random flips do not move the learned rule on average, they make it unstable. Under *one-sided* noise the picture depends on the model. For the shallow boosted models the damage is mainly bias (variance share 7% for AdaBoost and 17% for GB), because removing positives shifts the learned decision rule towards class 0. For the deep single tree and bagging it remains mainly variance (94% and 89%). Under *boundary* noise the damage is split roughly evenly (37–54% variance), and the share of bias grows with the noise rate. A variance share above 100% means that bias decreased slightly while variance increased.

C. Capacity Control: Ensemble Method or Tree Size?

Fig. 3 shows that deeper trees are damaged more in every family under symmetric and boundary noise (all $p < 0.03$). At equal depth the ordering of RQ1 reverses: depth-3 bagging and random forests are damaged 2.5–5.0 pp *less* than depth-3 boosting under symmetric and boundary noise (Table IV, all $p < 0.001$). This agrees with Dietterich [7], whose boosted trees were also deep, and shows that the apparent robustness of boosting in RQ1 is a tree-depth effect. Under one-sided noise tree depth makes no significant difference, consistent with its bias-type damage. Depth also changes the *type* of one-sided damage: its variance share rises from 17% for GB with depth-3 trees to 89% with depth-8 trees. Shallow bagging has, however, a higher clean error (18.0% against 13.6% for full-depth bagging), so being least damaged is not the same as being most accurate under noise.

TABLE IV
DAMAGE DIFFERENCE AT EQUAL DEPTH (PP, PARALLEL MINUS SEQUENTIAL, 20% NOISE). *: $p < 0.05$, WILCOXON

Parallel	Sequential	Sym.	One-s.	Bound.
Bagging_d3	AdaBoost_d3	−5.0*	−2.0*	−4.2*
Bagging_d3	GB	−3.4*	−1.4*	−3.1*
Bagging_d3	XGB_d3	−2.7*	−1.2*	−2.8*
RF_d3	AdaBoost_d3	−4.8*	0.5	−4.3*
RF_d3	GB	−3.2*	1.1	−3.2*
RF_d3	XGB_d3	−2.5*	1.3	−2.9*

D. RQ3: Where Does the Variance Come From?

Table V shows that the noise realisation and sampling add similar amounts of variance in most cases. Under boundary noise, however, the choice of flipped labels contributes more than resampling for bagging and random forests (9.6 vs. 7.5 pp and 8.1 vs. 5.8 pp). Averaging over bootstrap samples cannot remove this part, because every tree in the ensemble sees the same wrong labels. The boosting models have almost no variance of their own, whereas bagging and random forests have 2–4 pp from their internal randomness.

E. Where Does the Damage Land?

Table VI compares the damage in the near-boundary and far-from-boundary halves of the test sets. Boundary noise concentrates its damage near the boundary, 10 to 18 times more for the ensembles; one-sided noise does so moderately

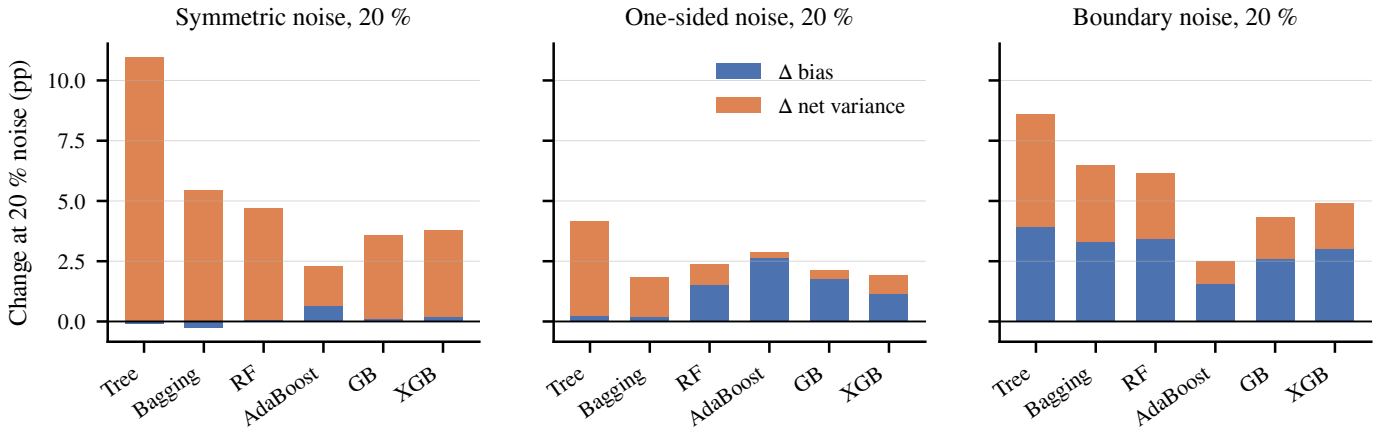


Fig. 2. Damage at 20% noise split into change in bias and change in net variance (averages over 15 datasets).

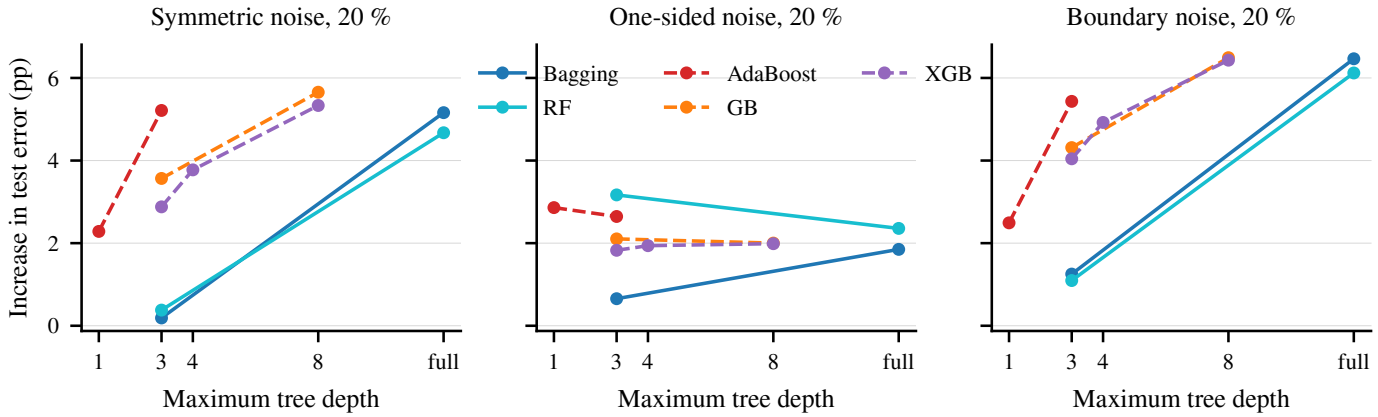


Fig. 3. Damage at 20% noise as a function of maximum tree depth for each ensemble family.

TABLE V
VARIANCE ADDED BY THE NOISE REALISATION AND BY SAMPLING (PP, 20% NOISE)

	Symmetric		One-sided		Boundary	
Model	Noise	Samp.	Noise	Samp.	Noise	Samp.
Tree	20.4	19.4	11.2	11.5	17.1	15.2
Bagging	9.0	9.0	4.1	5.2	9.6	7.5
RF	7.8	7.3	3.2	3.7	8.1	5.8
AdaBoost	8.3	9.1	4.3	5.3	7.8	7.9
GB	9.5	10.4	4.6	5.7	9.3	8.6
XGB	10.2	11.0	5.0	6.1	10.2	9.4

TABLE VI
RATIO OF DAMAGE NEAR VS. FAR FROM THE DECISION BOUNDARY (20% NOISE)

Model	Sym.	One-s.	Bound.
Tree	0.7	1.1	3.7
Bagging	1.0	1.8	10.2
RF	1.2	2.0	11.0
AdaBoost	1.5	3.8	17.9
GB	1.3	3.6	10.0
XGB	1.5	3.7	10.7

for boosting (about 3.7 times); and symmetric noise spreads it roughly evenly. For the single tree, symmetric noise even causes more damage far from the boundary (ratio 0.7), because a full-depth tree memorises flipped labels in otherwise confident regions.

F. RQ4: Do Common Remedies Help?

Table VII reports the share of the noise damage recovered by each remedy. Variance-reducing remedies work under symmetric and boundary noise: training bagging on 20% bootstrap samples recovers 56% and 33% of the damage (better on 14 of 15 datasets), and early stopping with a lower learning rate recovers about 19% for GB and XGB while also slightly lowering their clean error. Under one-sided noise none of the remedies helps significantly, and early stopping even increases the damage by about 11%. This is consistent with RQ2, since one-sided damage is mainly bias, which averaging and early stopping cannot correct, and with the theory of Long and Servedio [10]. The lower learning rate for AdaBoost reduces the damage but costs about 1 pp of clean accuracy, so its error under noise does not improve significantly.

TABLE VII
SHARE OF NOISE DAMAGE RECOVERED BY EACH REMEDY (20% NOISE). *: NOISY ERROR SIGNIFICANTLY LOWER, $p < 0.05$

Fix	Sym.	One-s.	Bound.
GB, early stop	19%*	−11%	7%*
XGB, early stop	19%*	−11%	8%*
AdaBoost, η=0.5	36%	14%	28%
Bagging, 20% samples	56%*	6%	33%*

TABLE VIII
MULTICLASS DATASETS: DAMAGE AT 20% NOISE (PP) AND SHARE DUE TO VARIANCE

	Symmetric		Pair-flip		Boundary	
Model	Δerr	Var.	Δerr	Var.	Δerr	Var.
Tree	13.0	103%	10.9	103%	8.4	70%
Bagging	1.8	131%	3.5	99%	4.1	59%
RF	1.9	104%	3.3	94%	4.2	55%
AdaBoost	−2.2	–	8.0	5%	1.8	56%
GB	3.3	106%	5.2	83%	4.1	52%
XGB	2.2	114%	4.3	75%	3.7	51%

G. RQ5: Do the Findings Hold for Multiclass Data?

Table VIII repeats the split of RQ2 on the 20 multiclass datasets. The two central findings carry over. Symmetric noise again adds almost only variance (variance share 103–131% for all models but AdaBoost), and boundary noise again splits roughly evenly (51–70% variance) with the bias share growing with the noise rate. For class-conditional (pair-flip) noise, stump-based AdaBoost is again almost pure bias (5%), whereas GB and XGB are now mostly variance (75–83%). This agrees with the capacity effect of Section V-C: class-conditional noise produces bias in low-capacity models and variance in higher-capacity ones. Unlike binary one-sided noise, the models differ significantly under pair-flip noise (Friedman $p < 10^{-6}$), and random forests and bagging have the best average damage ranks (2.2 and 2.6). Symmetric noise hurts the ensembles less than in the binary case (2–3 pp at 20%), because flips spread over many wrong classes rarely outvote the true class. AdaBoost with stumps fits multiclass data poorly (clean error 33% against about 20% for the other ensembles), and its error even falls slightly under symmetric noise, so its multiclass results should be read with caution.

VI. DISCUSSION

The results lead to a simple practical rule, summarised in Table IX: identify the likely type of label error from how the labels were produced, and choose the remedy accordingly. Random errors, such as data-entry slips, cause variance, which variance-reducing ensembles and regularisation can control. One-sided errors, such as under-reported positives, shift the decision rule; here the choice of ensemble does not matter significantly and regularisation does not help, so effort should go into correcting labels or re-weighting the affected class. Errors on ambiguous cases cause both kinds of damage and concentrate it on borderline test points, so both kinds of remedy are needed.

TABLE IX
PRACTICAL GUIDE DERIVED FROM THE RESULTS

Likely noise	Main damage	Model choice	Remedy
Random (symmetric)	Variance	Parallel ensemble with limited depth	Small-sample bagging, early stopping
One-sided	Bias (shallow), variance (deep)	No significant difference	Correct or re-weight labels; regularisation does not help
Ambiguous cases (boundary)	Bias and variance, near boundary	Parallel ensemble with limited depth	Variance reduction plus label review of borderline cases

The capacity experiment carries a methodological lesson: comparing ensemble families under default settings mixes the combination method with tree size. Under default settings boosting looked the most robust, but at equal depth the classic result of Dietterich [7] holds.

VII. LIMITATIONS

The main study uses binary tasks; the multiclass check covers fewer remedies and a single class-conditional pattern (pair-flip). All experiments use tree-based ensembles and synthetic label noise injected at controlled rates, whereas real noise may combine several types. Hyperparameters were fixed and not re-tuned for each noise level. Boundary noise was defined with a logistic-regression reference, which favours linear decision boundaries. The variance estimates use bootstrap samples from one training pool, which slightly understates the variance across independently drawn training sets.

VIII. CONCLUSION

Label noise does not damage all tree ensembles in the same way. Random noise mainly adds variance; one-sided noise adds bias to low-capacity models and variance to high-capacity ones; boundary noise adds both and concentrates near the decision boundary. At equal tree depth bagging and random forests are more robust than boosting, variance-reducing remedies help only when the damage is variance, and the main patterns also hold on multiclass data. Choosing the remedy according to the likely type of label error is therefore the key practical decision.

ACKNOWLEDGMENT

[TODO: Thank your guide/department, if appropriate.] [TODO: AI disclosure required by IEEE policy: name the AI tool used and state which parts it assisted with (e.g., code for the experiments, figure scripts, drafting or editing of specific sections), and confirm that the authors reviewed and take responsibility for all content.]

REFERENCES

[1] L. Breiman, "Bagging predictors," *Machine Learning*, vol. 24, no. 2, pp. 123–140, 1996.

[2] ——, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[3] Y. Freund and R. E. Schapire, "A decision-theoretic generalization of on-line learning and an application to boosting," *J. Comput. Syst. Sci.*, vol. 55, no. 1, pp. 119–139, 1997.

[4] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 785–794.

[6] B. Frénay and M. Verleysen, "Classification in the presence of label noise: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 5, pp. 845–869, 2014.

[7] T. G. Dietterich, "An experimental comparison of three methods for constructing ensembles of decision trees: Bagging, boosting, and randomization," *Machine Learning*, vol. 40, no. 2, pp. 139–157, 2000.

[8] Morampudi, Padi, Kakitapelli, and Surapani, "Noise-dependent robustness of XGBoost, LightGBM, and CatBoost," *Algorithms*, vol. 19, no. 9, p. 730, 2026, tODO-VERIFY author first names/initials from the article page.

[9] P. Domingos, "A unified bias-variance decomposition for zero-one and squared loss," in *Proc. 17th Nat. Conf. Artificial Intelligence (AAAI)*, 2000, tODO-VERIFY venue and pages against the PDF you use (AAAI-2000 vs ICML-2000 version).

[10] P. M. Long and R. A. Servedio, "Random classification noise defeats all convex potential boosters," *Machine Learning*, vol. 78, no. 3, pp. 287–304, 2010.

[11] N. Natarajan, I. S. Dhillon, P. K. Ravikumar, and A. Tewari, "Learning with noisy labels," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2013.

[12] J. A. Sáez and J. L. Romero-Béjar, "On the suitability of bagging-based ensembles with borderline label noise," *Mathematics*, vol. 10, no. 11, p. 1892, 2022.

[13] T. M. Khoshgoftaar, J. V. Hulse, and A. Napolitano, "Comparing boosting and bagging techniques with noisy and imbalanced data," *IEEE Trans. Syst., Man, Cybern. A, Syst., Humans*, vol. 41, no. 3, pp. 552–568, 2011.

[14] M. Sabzevari, G. Martínez-Muñoz, and A. Suárez, "Small margin ensembles can be robust to class-label noise," *Neurocomputing*, vol. 160, pp. 18–33, 2015.

[15] A. Eisenbürger, D. Otten, A. Hudde, and F. Hopfgartner, "Training gradient boosted decision trees on tabular data containing label noise for classification tasks," arXiv:2409.08647, 2024.

[16] D. Wood, T. Mu, A. M. Webb, H. W. J. Reeve, M. Luján, and G. Brown, "A unified theory of diversity in ensemble learning," *J. Mach. Learn. Res.*, vol. 24, 2023.

[17] He *et al.*, "Robust ensemble learning under label noise: A theoretical analysis and framework-specific solutions," *IEEE Trans. Neural Netw. Learn. Syst.*, 2026, early access. TODO-VERIFY full author list from IEEE Xplore.

[18] Mohammad, "How ensemble learning balances accuracy and overfitting: A bias-variance perspective on tabular data," arXiv:2512.05469, 2025, tODO-VERIFY author full name.

[19] J. Vanschoren, J. N. van Rijn, B. Bischl, and L. Torgo, "OpenML: Networked science in machine learning," *SIGKDD Explor.*, vol. 15, no. 2, pp. 49–60, 2014.

[20] B. Bischl *et al.*, "OpenML benchmarking suites," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021, tODO-VERIFY full author list.

[21] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[22] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, 2006.